

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T43 · Area and Perimeter

With Answers

Questions with full worked solutions

■ 34 pages

FAMILY MAP

Area Perimeter

F01	Use parallelogram and trapezium area	5 Q 19 marks
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F02	Use rectangle and triangle area	7 Q 32 marks
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F03	Find perimeter of polygonal shapes	3 Q 11 marks
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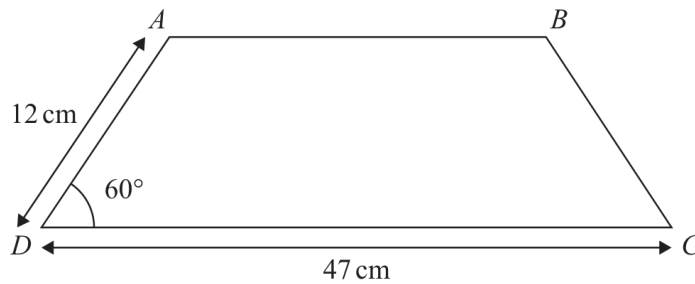
Working Unit 06

4MA1-1H Q9

ORIGINAL QUESTION

5 marks

9

Diagram **NOT**
accurately drawn

$ABCD$ is a trapezium with one line of symmetry.

$$\text{angle } ADC = 60^\circ \quad AD = 12 \text{ cm} \quad DC = 47 \text{ cm}$$

Work out the area of the trapezium.
Give your answer correct to 3 significant figures.
Show your working clearly.

..... cm^2

(Total for Question 9 is 5 marks)

Worked Solution 06

Area of a symmetric trapezium

UNIT 1
5 marks**Height and horizontal projection**

$$h = 12 \sin 60^\circ = 6\sqrt{3}, \quad 12 \cos 60^\circ = 6.$$

Upper base

$$AB = 47 - 6 - 6 = 35 \text{ cm.}$$

Area

$$A = \frac{1}{2}(47 + 35)(6\sqrt{3}) = 246\sqrt{3} = 426.084 \dots$$

Final answer

$$426 \text{ cm}^2 \text{ (3 s.f.)}$$

Dependency check The line of symmetry makes the two horizontal projections equal.

Question 21

4MA1-2H Q7

ORIGINAL QUESTION

4 marks

- 7 The diagram shows a 6-sided shape $ABCDEF$

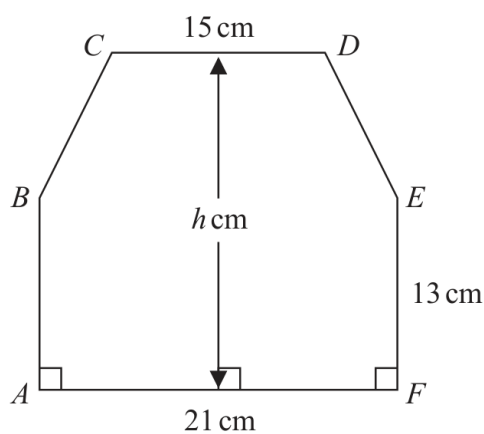


Diagram **NOT**
accurately drawn

$$AF = 21 \text{ cm} \quad CD = 15 \text{ cm} \quad AB = FE = 13 \text{ cm}$$

The perpendicular height of the shape is h cm

CD is parallel to AF

The area of the shape is 390 cm^2

Work out the value of h

$h = \dots\dots\dots$

Worked Solution 21

Composite area

MODULAR UNIT 1

4 marks

Area equation

$$21(13) + \frac{1}{2}(21 + 15)(h - 13) = 390.$$

Solve

$$273 + 18(h - 13) = 390 \Rightarrow h = 19.5.$$

Final answer

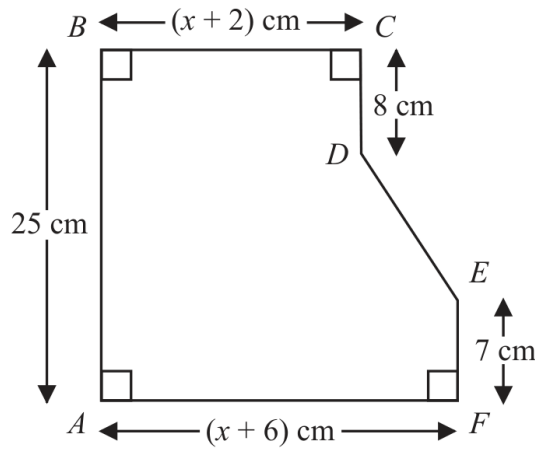
19.5 cm

Question 22

4MA1-2H Q10 whole

ORIGINAL QUESTION UNIT 1

5 marks

10 The diagram shows a hexagon $ABCDEF$ Diagram **NOT**
accurately drawn

$$AB = 25 \text{ cm} \quad BC = (x + 2) \text{ cm} \quad CD = 8 \text{ cm} \quad EF = 7 \text{ cm} \quad AF = (x + 6) \text{ cm}$$

The area of hexagon $ABCDEF$ is 258 cm^2 Work out the value of x $x = \dots\dots\dots$

Worked solution 22

Area of a stepped hexagon

MODULAR UNIT 1

5 marks

whole – Area of a stepped hexagon

Read the missing vertical lengths

DE drops $25 - 8 - 7 = 10$ cm; the horizontal width of the sloping strip is $(x + 6) - (x + 2) = 4$ cm.

Split into a rectangle and trapezium

$$A = 25(x + 2) + \frac{1}{2}(17 + 7)(4) = 25(x + 2) + 48.$$

Use the given area

$$25(x + 2) + 48 = 258 \Rightarrow 25x + 98 = 258.$$

Solve

$$25x = 160 \Rightarrow x = 6.4.$$

Final answer

$$x = 6.4$$

Original question 18

4MA1-2H Q5 Q5

ORIGINAL QUESTION**4 marks**

- 5 The diagram shows an 8-sided shape $ABCDEFGH$.

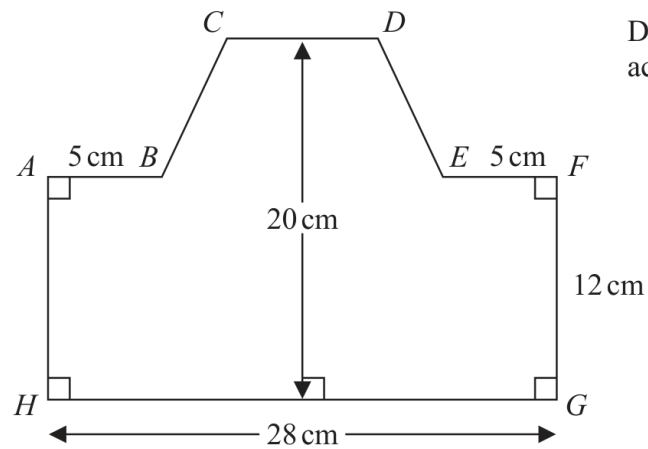


Diagram **NOT**
accurately drawn

$$HG = 28 \text{ cm} \quad FG = 12 \text{ cm} \quad AB = EF = 5 \text{ cm}$$

The height of the shape is 20 cm

CD is parallel to HG

The area of shape $ABCDEFGH$ is 434 cm^2

Find the length of CD .

..... cm

Worked solution

Family: Use parallelogram and trapezium area • Variant: Recover a missing trapezium or parallelogram

METHOD / WORKING

MODULAR UNIT 1 • JUNE 2022 DETAILED SOLUTIONS
Worked solution 18 MODULAR UNIT 1
Chapter: Area & Perimeter 4 marks
Q5 – Chapter: Area & Perimeter
Family: Use parallelogram and trapezium area • Variant: Recover a missing trapezium or parallelogram
dimension
Method
<i>The lower rectangle has area $28 \times 12 = 336 \text{ cm}^2$.</i>
Why: Its width and height are given by HG and FG.
Method
The top section has height $20 - 12 = 8 \text{ cm}$ and lower parallel side $28 - 5 - 5 = 18 \text{ cm}$.
Why: Remove the two 5 cm horizontal offsets.
Method
If $CD = x$, then $336 + (1/2)(18 + x) \times 8 = 434$.
Why: Add the rectangle area to the trapezium area.
Method
$4(18 + x) = 98$, so $x = 6.5$ and $CD = 6.5 \text{ cm}$.
Why: Solve the resulting linear equation.
Final answer
CD: $CD = 6.5 \text{ cm}$

FINAL ANSWER

Original question 07

4MA1-1H Q9

ORIGINAL QUESTION**4 marks**

- 9 The diagram shows isosceles triangle ABC

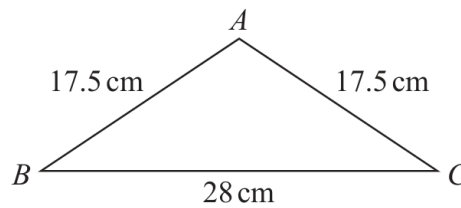


Diagram **NOT**
accurately drawn

$$AB = AC = 17.5 \text{ cm}$$

$$BC = 28 \text{ cm}$$

Calculate the area of triangle ABC

..... cm^2

Worked solution

Family: Use parallelogram and trapezium area • Variant: Combine polygon area with Pythagoras or

METHOD / WORKING

MODULAR UNIT 1 • JANUARY 2022 DETAILED SOLUTIONS
Worked solution 07 MODULAR UNIT 1
Chapter: Area & Perimeter 4 marks
Q9 – Chapter: Area & Perimeter
Family: Use parallelogram and trapezium area • Variant: Combine polygon area with Pythagoras or
trigonometry
Method
The perpendicular height is $\sqrt{(17.5^2 - 14^2)} = \sqrt{110.25} = 10.5$ cm.
Why: Bisect the 28 cm base of the isosceles triangle and use Pythagoras.
Method
Area = $\frac{1}{2} \times 28 \times 10.5$.
Why: Use base times perpendicular height divided by 2.
Method
$Area = 147cm^2$.
Why: Evaluate the product.
Method
<i>Therefore the area is $147cm^2$.</i>
Why: State the requested area.
Final answer
<i>area : $147cm^2$</i>

FINAL ANSWER

- 3 The diagram shows a shape.

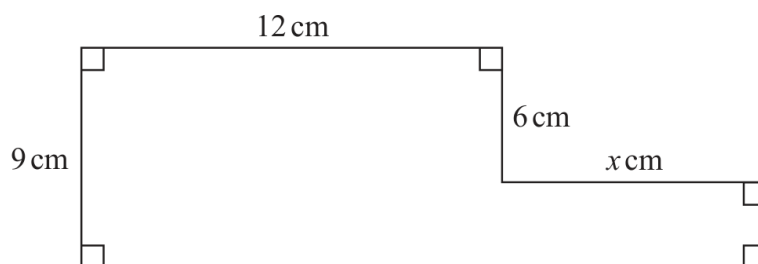


Diagram **NOT**
accurately drawn

The shape has area 129 cm^2

Work out the value of x .

$x =$

Worked solution

January 2020 | Question 3 | 4 marks

Family: Use rectangle and triangle area • Variant: Recover a missing dimension from area

Method / working

1. The left rectangle has area $12 \times 9 = 108$.
2. The right strip has height $9 - 6 = 3$ and area $3x$.
3. $108 + 3x = 129$, so $3x = 21$.
4. Therefore $x = 7$ cm.

Final answer: $x = 7$ cm

14 The diagram shows parallelogram $EFGH$.

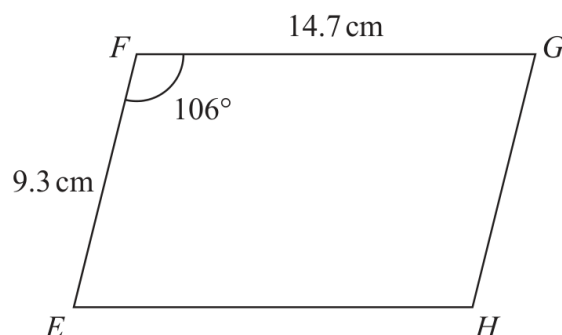


Diagram **NOT**
accurately drawn

$$EF = 9.3 \text{ cm}$$

$$FG = 14.7 \text{ cm}$$

$$\text{Angle } EFG = 106^\circ$$

- (a) Work out the area of the parallelogram.
Give your answer correct to 3 significant figures.

..... cm^2
(2)

Worked solution

January 2020 | Question 14 | 2 marks

Family: Use parallelogram and trapezium area | Variant: Find area of a parallelogram

METHOD / WORKING

$$1. \text{Area} = ab \sin \theta = 9.3 \times 14.7 \times \sin 106^\circ.$$

$$2. \text{Area} = 131.414... \text{ } 131\text{cm}^2 (3\text{s.f.}).$$

FINAL ANSWER

$$131\text{cm}^2$$

18 A rectangle $ABCD$ is to be drawn on a centimetre grid such that

A has coordinates $(-4, -2)$

B has coordinates $(1, 10)$

C has coordinates $(19, a)$

D has coordinates (b, c)

(a) Work out the value of a , the value of b and the value of c .

$a = \dots\dots\dots$

$b = \dots\dots\dots$

$c = \dots\dots\dots$

(4)

(b) Calculate the perimeter, in centimetres, of rectangle $ABCD$.

$\dots\dots\dots$ cm

(3)

Worked solution

January 2021 | Question 18 | 3 marks

Family: Find perimeter of polygonal shapes • Variant: Find perimeter of a simple triangle or rectangle

Method / working

1. $AB = \sqrt{5^2 + 12^2} = 13 \text{ cm.}$
2. $BC = \sqrt{18^2 + 7.5^2} = 19.5 \text{ cm.}$
3. $\text{Perimeter} = 2(13 + 19.5) = 65 \text{ cm.}$

Final answer: 65 cm

20

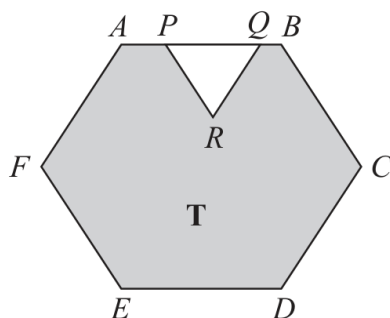


Diagram **NOT**
accurately drawn

The diagram shows a shaded region **T** formed by removing an equilateral triangle PQR from a regular hexagon $ABCDEF$.

The points P and Q lie on AB such that $AB = 1.5 \times PQ$

Given that the area of region **T** is $72\sqrt{3} \text{ cm}^2$

work out the length of PQ .

..... cm

Worked solution

January 2021 | Question 20 | 4 marks

Family: Use rectangle and triangle area | Variant: Find composite or shaded polygonal area

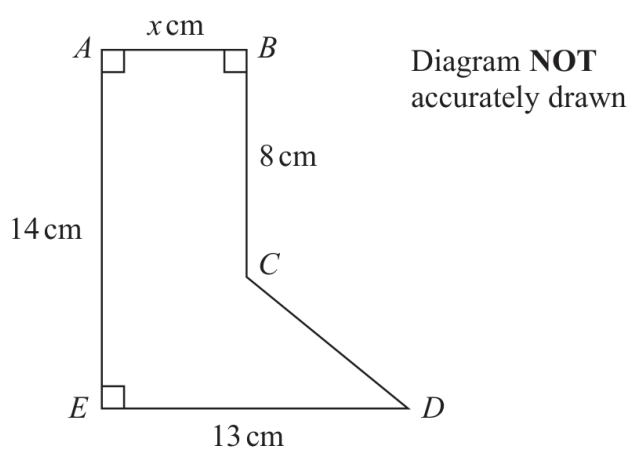
METHOD / WORKING

1. Let $PQ=x$, so $AB=1.5x$.
2. Write the equilateral triangle and regular-hexagon areas in terms of x .
3. *Set shaded area equal* to $72\sqrt{3}$ and solve the resulting quadratic in x^2 .
4. The positive length is $PQ=4.8$ cm.

FINAL ANSWER

4.8 cm

3



The diagram shows the shape $ABCDE$.

The area of the shape is 91.8 cm^2

Work out the value of x .

$x = \dots\dots\dots$

Worked solution

June 2021 | Question 3 | 4 marks

Family: Use rectangle and triangle area • Variant: Recover a missing dimension from area

Method / working

1. Write the area equation for the composite shape: $11x + 39 = 91.8$.
2. Subtract 39: $11x = 52.8$.
3. Divide by 11 to get $x = 4.8$.
4. Check the positive length and give cm.

Final answer: 4.8 cm

- 4 The diagram shows the front of a wooden door with a semicircular glass window.

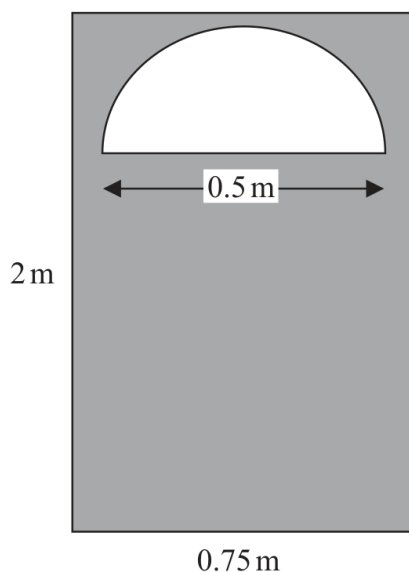


Diagram **NOT**
accurately drawn

Julie wants to apply 2 coats of wood varnish to the front of the door, shown shaded in the diagram.

250 millilitres of wood varnish covers 4 m^2 of the wood.

Work out how many millilitres of wood varnish Julie will need.
Give your answer correct to the nearest millilitre.

..... millilitres

Worked solution

November 2021 | Question 4 | 5 marks

Family: Use rectangle and triangle area | Variant: Use an area result in a real-world quantity or cost

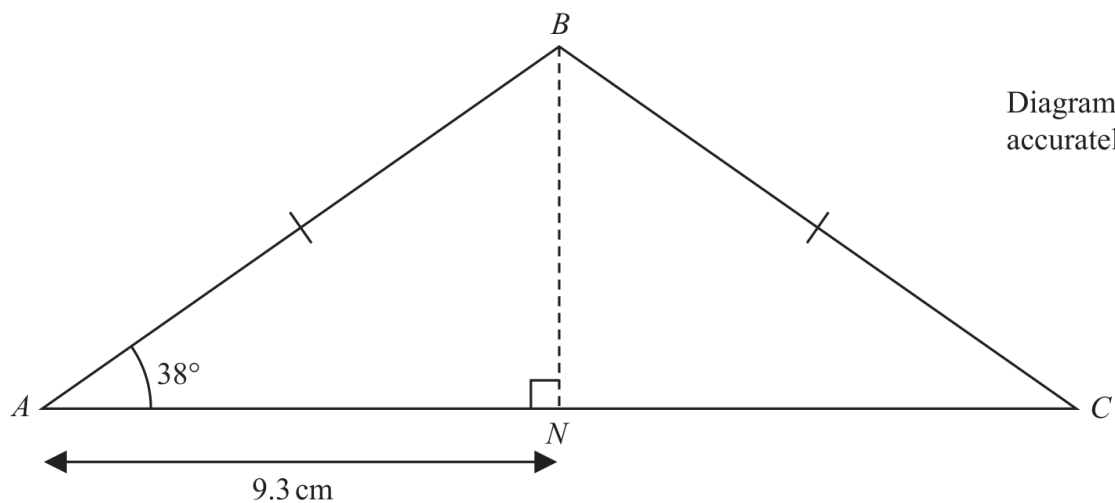
METHOD / WORKING

1. Area of the rectangular door:
2. $0.75 \times 2 = 1.5m^2$
3. The semicircular window has diameter $0.5m$, so
4. $r=0.25$ m
5. Area of the semicircular window:
6. $\frac{1}{2} \pi (0.25)^2 = 0.098174...m^2$
7. Shaded area for one coat:
8. $1.5 - 0.098174... = 1.401825...m^2$
9. For 2 coats:
10. $2 \times 1.401825... = 2.80365...m^2$
11. Since 250 ml covers $4m^2$, varnish needed is
12. $\frac{2.80365...}{4} \times 250 = 175.22...$
13. Answer: 175 millilitres.

FINAL ANSWER

175 millilitres.

10 ABC is an isosceles triangle with $BA = BC$.



N is the point on AC such that $AN = 9.3$ cm and BN is perpendicular to AC .

Work out the perimeter of triangle ABC .

Give your answer correct to 3 significant figures.

..... cm

Worked solution

November 2021 | Question 10 | 4 marks

Family: Find perimeter of polygonal shapes | Variant: Find perimeter of a composite polygonal shape

METHOD / WORKING

1. In right-angled triangle ABN ,
2. $\cos 38^\circ = \frac{9.3}{AB}$
3. $AB = \frac{9.3}{\cos 38^\circ} = 11.802\dots$
4. Since ABC is isosceles,
5. $BC=AB$
6. Also,
7. $AC = 2 \times 9.3 = 18.6$
8. Perimeter:
9. $11.802\dots + 11.802\dots + 18.6 = 42.204\dots$
10. Answer: 42.2cm .

FINAL ANSWER

42.2 cm.

- 11 The diagram shows trapezium $ABCD$ in which BC and AD are parallel.

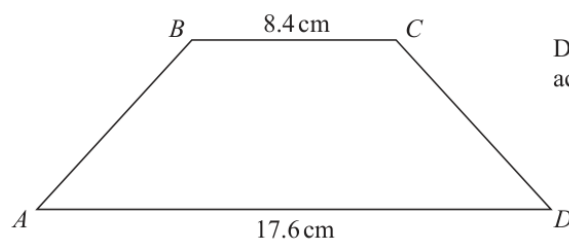


Diagram **NOT**
accurately drawn

The trapezium has exactly one line of symmetry.

$$BC = 8.4 \text{ cm}$$

$$AD = 17.6 \text{ cm}$$

The trapezium has area 179.4 cm^2

Work out the size of angle ABC .

Give your answer correct to 1 decimal place.

(Total for Question 11 is 6 marks)

Answer reference | May-Nov 2020 | Question 11

Family: Use rectangle and triangle area • Variant: Recover a missing dimension from area

11	$(AX \Rightarrow) (17.6 - 8.4) \div 2 (= 4.6)$		6	M1 where X is the foot of the perpendicular from B to AD
	$0.5 \times (8.4 + 17.6) \times h = 179.4$ or $0.5 \times '4.6' \times h + 0.5 \times '4.6' \times h + 8.4 \times h = 179.4$ or $13 \times h = 179.4$			M1
	$(h \Rightarrow) 179.4 \div '13' (=13.8)$ or $(h \Rightarrow) 358.8 \div '26' (=13.8)$ oe			M1
	$\tan ABX = \frac{'4.6'}{'13.8'}$ or $\tan BAX = \frac{'13.8'}{'4.6'}$			M1 ft their h dep on second M1 $(AB \Rightarrow) \sqrt{'4.6'^2 + '13.8'^2} = \sqrt{211.6}$ $= (14.546\dots)$ and one from $\sin ABX = \frac{'4.6'}{\sqrt{211.6}}$ or $\sin BAX = \frac{'13.8'}{\sqrt{211.6}}$ or $\cos ABX = \frac{'13.8'}{\sqrt{211.6}}$ or $\cos BAX = \frac{'4.6'}{\sqrt{211.6}}$ or $\sin ABX = \frac{'4.6' \times \sin 90}{\sqrt{211.6}}$ or $\cos ABX = \frac{\sqrt{211.6} \times '13.8'^2 - '4.6'^2}{2 \times \sqrt{211.6} \times '13.8'}$
	$(ABX \Rightarrow) \tan^{-1} \left(\frac{'4.6'}{'13.8'} \right) (= 18.4)$ or $(BAX \Rightarrow) \tan^{-1} \left(\frac{'13.8'}{'4.6'} \right) (= 71.6)$			M1
		108.4		A1 awrt 108.4
				Total 6 marks

Worked solution

May-Nov 2020 | Question 11 | 6 marks

Family: Use rectangle and triangle area | Variant: Recover a missing dimension from area

METHOD / WORKING

Worked solution
May-Nov 2020 Question 11 6 marks
Family: Use rectangle and triangle area Variant: Recover a missing dimension from area
Solution
The average of the parallel sides is
$(8.4+17.6)/(2)=13$
So the perpendicular height is
$179.4 / 13=13.8$
The horizontal offset on each side is
$(17.6-8.4)/(2)=4.6$
The acute angle between AB and the horizontal is
$\tan^{-1}((13.8)/(4.6)) = 71.565\dots^{\circ}$
So the interior angle ABC is
$180^{\circ} - 71.565\dots^{\circ} = 108.434\dots^{\circ}$
Answer: 108.4°

FINAL ANSWER

- 8 The diagram shows an isosceles triangle.

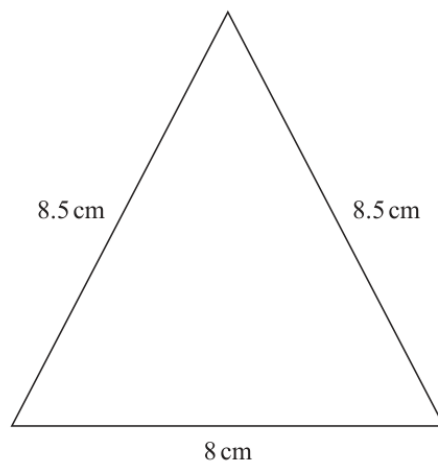


Diagram **NOT**
accurately drawn

Work out the area of the triangle.

.....cm²

(Total for Question 8 is 4 marks)

Answer reference | May-Nov 2020 | Question 8

Family: Use rectangle and triangle area • Variant: Recover a missing dimension from area

8	$8.5^2 - (8 \div 2)^2 (= 56.25)$ or $\cos x = \frac{4}{8.5}$ oe			M1 or eg $\cos A = \frac{8^2 + 8.5^2 - 8.5^2}{2 \times 8 \times 8.5}$
	$\sqrt{56.25}$ (= 7.5) or $x = \cos^{-1}\left(\frac{4}{8.5}\right)$ (= 61.927...) oe			M1 or eg $(A =) \cos^{-1}\left(\frac{8^2 + 8.5^2 - 8.5^2}{2 \times 8 \times 8.5}\right)$ (61.927...) (other angle = 56.144...)
	$8 \times "7.5" \div 2$ oe or $0.5 \times 8 \times 8.5 \times \sin "61.927..."$			M1 or eg $0.5 \times 8.5 \times 8 \times \sin "61.927..."$ oe
		30	4	A1
				Total 4 marks

Worked solution

May-Nov 2020 | Question 8 | 4 marks

Family: Use rectangle and triangle area • Variant: Recover a missing dimension from area

Official final mark-scheme pages are embedded immediately after this question.

- 4 The diagram shows rectangle $ABCD$

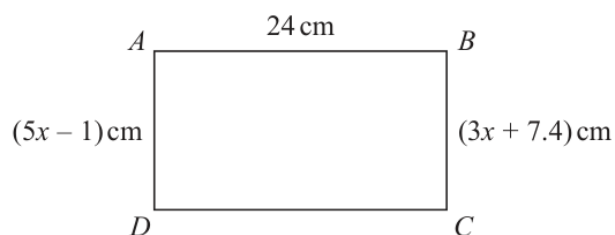


Diagram **NOT**
accurately drawn

Work out the perimeter of the rectangle.
Show your working clearly.

..... cm

(Total for Question 4 is 4 marks)

Answer reference | January 2023 | Question 4

Family: Find perimeter of polygonal shapes • Variant: Find perimeter of a simple triangle or rectangle

Question	working	Answer	Mark	Notes
4	eg $5x - 1 = 3x + 7.4$ oe or eg $10x - 2 + 48$ or $6x + 14.8 + 48$ or $24 + 24 + 5x - 1 + 3x + 7.4$ oe		4	M1 a correct equation to find x or a correct expression for the perimeter in terms of x
	$x = 4.2$			A1 the correct value of x (implies previous mark)
	$2 \times 24 + 2(5 \times "4.2" - 1)$ oe or $2 \times 24 + 2(3 \times "4.2" + 7.4)$ oe or $2 \times 24 + (5 \times 4.2 - 1) + (3 \times 4.2 + 7.4)$ oe eg $24 + 24 + 20 + 20$ oe			M1dep on a correct method to find the perimeter – use of positive x from correct working (1 st M1 awarded for an equation) and only if used the same measurement for AD and BC
	working required	88		A1 cao dep on either M1 or $x = 4.2$
				Total 4 marks

Worked solution

January 2023 | Question 4 | 4 marks

Family: Find perimeter of polygonal shapes • Variant: Find perimeter of a simple triangle or rectangle

Official final mark-scheme pages are embedded immediately after this question.